

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 35

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Domesticated crops were bred from a small number of original plants so they have a small gene pool / low genetic diversity • Selective breeding for yield, appearance or shelf life may have lost other useful genes (eg taste, disease resistance) • Genetically similar plants share the same susceptibilities so a new disease or environmental change can affect the whole crop	3
01.2	• A region with a high concentration of the wild relatives of important crops - usually where the crop was first domesticated (eg wheat / barley in the Middle East; potato, tomato in South America; rice in India; maize in Central America) • They hold the largest gene pools / most genetic variety for future crop breeding (disease resistance, drought and salt tolerance, yield) • Many are in regions where environmental degradation / habitat loss threatens the wild plants	3
01.3	• A large population may all descend from a few closely related individuals so they are genetically similar • Similar adaptations mean all are susceptible to the same change (disease, climate) • Greater risk of inbreeding so disadvantageous recessive genes are expressed (eg amelanism, poor muscle development in lion cubs); so species should be protected across their whole range to retain regional genes	3
01.4	• Morama bean (drought resistant, soya-like protein); yeheb tree (drought resistant nuts, poor soils); potato bean (high protein); perennial maize / Kernza (no annual replanting, less ploughing / erosion); salt-tolerant spinach relatives	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 15-17; AQA spec 3.1.2.1.2.	

Question	Marking guidance	Marks
02.1	• Small clearings were farmed for a few years then abandoned as nutrients declined; the forest recolonised and soil fertility recovered before the area was cleared again • Population growth shortens the gap between clearances so the forest and soil do not recover • Clearings enlarge and degrade; less regrowth	3
02.2	• Rainforest soils are thin and nutrient-poor (nutrients held in the biomass); once cleared, fertility falls in a few years and yields drop, so more land is cleared • Abandoned land may not be recolonised if no forest survives nearby; erosion of exposed soil	2
02.3	• Mining (aluminium, iron, gold, copper) destroys habitat directly and pollutes rivers with turbid drainage, cyanide and mercury from processing • Reservoirs (eg Tucurui, 2850 km²) flood large areas, block migratory fish and river dolphins; decomposing vegetation produces acidic, deoxygenated water; fluctuating downstream flows • Droughts increase frequency, area and duration of fires (long fires boil sap and kill trees); cloud forests lose humidity as clouds form higher so frogs and epiphytes decline; some models predict replacement by savannah	3
02.4	• A creditor country or conservation charity cancels / buys part of a debtor country's foreign debt • In return the country guarantees to protect an area of rainforest (eg USA - Guatemala, Brazil; WWF - Costa Rica, Philippines)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 55-57; AQA spec 3.1.2.3.4.	
03.1	• Anchovy r-selected; blue whale k-selected • r-selected: mature quickly, produce many young, disperse widely, short lives; adapted to respond rapidly to high death rates / unstable conditions (mice, locusts, greenfly, rabbits) • k-selected: mature late, few young with parental care, long-lived; populations stable near carrying capacity (whales, elephants, rhinos)	3
03.2	• Low reproductive rate means losses from hunting / habitat change cannot be replaced quickly • An increase in death rate above the low birth rate causes a population crash • Recovery takes decades (eg blue whale still far below pre-whaling numbers after the 1986 moratorium); r-selected species rebound within a few years once exploitation stops	3
03.3	• New population = starting population + births + immigrants - deaths - emigrants	1
03.4	• The greatest rate of exploitation (harvest) possible without causing long-term population decline • Equals the surplus production: highest when the population is well below carrying capacity where growth rate is fastest (middle of the sigmoid curve) • At carrying capacity net growth is zero so any harvest reduces the population; harvesting above MSY causes decline	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 79-80; AQA spec 3.1.3.4.	

Question	Marking guidance	Marks
04.1	• B: frost-free period (150 days) exceeds the 120 needed whereas A (110) is too short - late frosts would kill seedlings • B is warmer / longer thermal growing season because south-facing slopes receive more insolation; valley floors are frost pockets where cold dense air collects • But B has lower rainfall and a steeper slope (erosion / machinery difficulties) so irrigation / contour cultivation may be needed	3
04.2	<i>Any 3 from:</i> • Greenhouses / polytunnels heated by solar gain or burning gas / paraffin (ventilated to prevent overheating) • Woven plastic cloches / fleece over rows to retain warm air near the ground • Burning oil / gas or using large fans in orchards to prevent frost damage to blossom • Site selection (aspect, avoiding frost pockets); heated buildings and shelter for livestock	3
04.3	• Mammals and birds use food energy to maintain a constant body temperature (thermoregulation) • Reducing heat loss (shelter, heated buildings) means less food energy is used for heat and more is available for growth	2
04.4	• Long days increase milk production and egg laying; poultry grow best with short days; sheep mate as days shorten in autumn • Artificial lighting extends the growing season for greenhouse crops or simulates autumn in spring to give sheep a second mating / lambing season	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 306-309; AQA spec 3.5.1.1.	
05.1	• Image databases: identifying individuals (tiger stripes, whale shark spots, dolphin fins) to find territory size, movements, lifespan, social groups - few species have unique recognisable features • eDNA: detecting the presence of a species from DNA shed into water (great crested newt, invasive carp) - shows presence not abundance; contamination / DNA transport by water • Geolocator tags: tracking small birds on migration where transmitters are too heavy - only record light and time so position is approximate; animal must be recaptured to download data	6
05.2	• Microwaves emitted by the satellite reflect off the surface giving data on wave height, wind velocity, sea surface altitude and oil slicks (which smooth the surface) • Microwaves pass through cloud so monitoring is not interrupted by weather	2
05.3	<i>Any 2 from:</i> • Droppings: diet, gender, territory size • Owl pellets: diet • Feeding marks (chewed hazel nuts - dormouse), tracks / footprints, nests / burrows, scratching posts (territory)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402-407; AQA spec 3.7.2.3.	

Question	Marking guidance	Marks
06.1	<i>Any 3 from:</i> • Sinkers / weighted lines so hooks sink quickly below the reach of albatrosses • Setting lines at night when birds are not feeding • Bird-scaring streamers / decoys • Curved hooks that catch tuna but not birds	3
06.2	• Effort: limits on boat size, engine power, days at sea, number of licences; decommissioning payments to reduce fleet size where overfishing is unavoidable • Methods: bans or length limits on drift nets; bans on demersal trawling over sensitive seabed (Rockall Bank, Hatton Bank deep-water reefs) • Shorter trawl tows so turtles / by-catch survive (over 99% survive under 10 minutes)	3
06.3	• Eggs / larvae are reared in captivity through the planktonic stage when most die (predation, failure to find habitat) • Juveniles are released when survival chances are good, increasing recruitment to the adult population	2
06.4	• Egg / larval survival varies greatly between years so adult numbers do not predict recruitment • Several poor years would lower the MSY; egg surveys allow catches to be reduced in advance rather than after a sudden collapse	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 344-348; AQA spec 3.5.2.2.	
07.1	• Tracking: continuous data on movements, territory size, migration routes and habitat use without disturbance, over large areas and long periods; acoustic tags in seawater; geolocators for small birds; more representative than occasional sightings • DNA: identifies individuals, gene pool size / inbreeding and regional populations (tracing ivory, timber, fish); eDNA detects rare or hidden aquatic species (great crested newt, invasive carp) from shed cells without capture • Image recognition: software identifies individuals from unique markings (tiger stripes, whale shark spots, fin damage) in camera-trap / drone / satellite images, giving population size, lifespan, social structure from large image sets quickly and objectively • Acoustic: bat detectors, hydrophones and sonograms record species that are hard to see (bats, cetaceans, insects), giving presence, abundance and activity continuously, day and night	8
07.2	• Data collected continuously / over large areas / at night rather than at a few times and places a researcher can visit • Less disturbance so animals behave naturally; larger sample sizes; objective rather than subjective identification	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402-406; AQA spec 3.1.2.3.6.	
08.1	• Moral belief: protecting the environment is the 'right' thing to do regardless of benefit to people • Enlightened self-interest: protecting the environment is done because it causes fewer difficulties (for people) than not doing so	2

Question	Marking guidance	Marks
08.2	• Total impact = population size x the individual (per capita) impact of each person • Per capita impact rises with affluence / access to energy, so both a larger population and a more affluent population increase the total impact	2
08.3	<i>Any 1 from:</i> • Slash and burn agriculture • Fishing • Discharge of sewage / biodegradable pollutants	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370; AQA spec 3.6.1.	
09.1	• Yield increases as inputs increase • But each extra unit of input produces a smaller increase in yield than the previous one (the curve flattens / law of diminishing returns)	2
09.2	• Second unit: $70 - 50 = 20$ units • Fifth unit: $94 - 90 = 4$ units	2
09.3	• The first units of input on each area give the largest yield increases • Concentrating inputs means later units add very little, whereas the same inputs spread thinly all work at the steep part of the curve	2
09.4	• Some areas have much greater potential productivity so investment there is sensible • Affluent societies can afford to buy more inputs while poorer communities that would benefit most cannot afford them (economic reasons)	2
09.5	• Yield per unit area is very high because of large inputs • But the extra yield per unit of input is small, so efficiency (yield per unit input / energy ratio) is low	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 324-325; AQA spec 3.5.1.2.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • In-situ: species remain in natural conditions with the whole community and inter-species relationships; protected areas (SSSI, NNR, SAC, SPA, Ramsar, MCZ), land ownership, agri-environment schemes, habitat creation and design (area, shape, corridors, diversity), management of abiotic and biotic factors, plagioclimax maintenance, population control, reintroductions and rewilding (Yellowstone wolves, Scottish beavers) • Limits of in-situ: external threats (pollution, climate change), enforcement, cost, conflicts with land use, small fragmented populations • Ex-situ: captive breeding when in-situ is impossible or insufficient; selection criteria; difficulties keeping and breeding (habitat size, food, relationships, breeding triggers, group dynamics, gene pool / inbreeding, hybridisation); techniques (stud books, cryopreservation, AI, embryo transfer, micropropagation, cloning); seed banks; successes (Hawaiian goose, Iberian lynx, red kite, large blue) • Limits of ex-situ: expensive, few species, loss of wild behaviour, small gene pools, release requires suitable habitat and post-release support (hard vs soft release, monitoring) • Judgement: ex-situ is a temporary safeguard / support that only succeeds if in-situ habitat exists to return to; both needed, in-situ fundamental	9
10.2	• Restoring natural processes / communities to a habitat changed by humans, often by reintroducing missing (keystone) species and removing human management such as grazing	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 32-49; AQA spec 3.1.2.3.3-3.1.2.3.4.	

Question	Marking guidance	Marks
11.1	Discuss the extent to which the resources and services provided by other species justify the conservation of biodiversity. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Resources: wood, fibres, oils, fuels, food; new food species (morama bean, yeheb, perennial maize, Kernza, bison, eland) and their adaptations • Biomimetics: wing-tip feathers, whale tubercles, shark skin, termite mounds, bird bones, gecko adhesion, burrs, lotus, spider silk • Medicines: poppies, cinchona, willow, yew (taxol), Mexican yam, sponges (AZT), sharks; few species yet studied; physiological research (marsupials, hippo, sponges) • Genetic resources: crop wild relatives, Vavilov centres, gene pools and inbreeding • Ecosystem services: atmospheric regulation, hydrological cycle, biogeochemical cycles, soil maintenance; interspecies relationships - food, pollination (Darwin's orchid), seed dispersal, habitat provision • Evaluation: economic vs intrinsic / moral arguments; services unnoticed because of dynamic equilibrium; unknown future value; justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 10-19; AQA spec 3.1.2.1.	25

Question	Marking guidance	Marks
11.2	Discuss how feedback mechanisms, tipping points and diversity influence the stability of natural systems, and how human activities affect them. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Sustainability definitions; population and per capita impact; attitudes of rural and urban societies • Negative feedback: cloud cover and albedo, CO₂ and photosynthesis, evaporation and precipitation, density-dependent population regulation • Positive feedback driven by human activities: permafrost, acidification, albedo decline, methane hydrates, fires, cirrus clouds, decomposition • Tipping points: rare under negative feedback; large events (Permian methane hydrate); self-sustaining change from human activities • Diversity and resistance: reefs and rainforests vs low-diversity agroecosystems lacking detritivores, pollinators and predators • Human responses that override feedback (fertilisers, irrigation, pesticides); links to climate change chapter; justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-372; AQA spec 3.6.1.	25